

NETFLIX PRIZE

Recommendation System

Personalized Content Discovery using Collaborative Filtering & Matrix Factorization

Python 3

scikit-surprise

SVD

Item-Based
CF

MAP@10

RMSE

By-

Krish Kumar (24115088)
Mohd Saif (24115104)

The Challenge

- 100M+ ratings across 480K users and 17,770 movies
- Dataset sparsity of ~98.8% — users rate very few movies
- Predict unseen ratings and rank recommendations accurately
- Balance prediction accuracy (RMSE) vs ranking quality (MAP@10)

17,770

Movies

1-5

Star Scale

100M+

Ratings

480K

Users

Key Insights from EDA

Movie Popularity

Long tail — most movies have very few ratings

Data Sparsity

98.8% sparse matrix;
latent factor models preferred

Rating Distribution

Left-skewed — 60% of ratings are 4 or 5 stars (positivity bias)

User Activity

Power-law distribution;
top 1% users = 10-15% of ratings

Relevance threshold set at 3.5 stars — separating liked (4-5) from disliked (1-3) content

Recommendation Pipeline

1

Raw Data
Parsing

Netflix .txt
format parser

2

Sampling &
Filtering

2M rows;
min 50/100 ratings

3

Train-Test
Split

80/20 split
random_state=42

4

Model
Training

IBCF + SVD
GridSearch CV

5

Evaluation

RMSE, MAE
MAP@10

Item-Based CF

KNNWithMeans | k=40 | Pearson similarity

RMSE: 0.9720

BEST

n_factors=100 | n_epochs=30 | GridSearch tuned

RMSE: 0.9273

Item-Based CF vs SVD Matrix Factorization

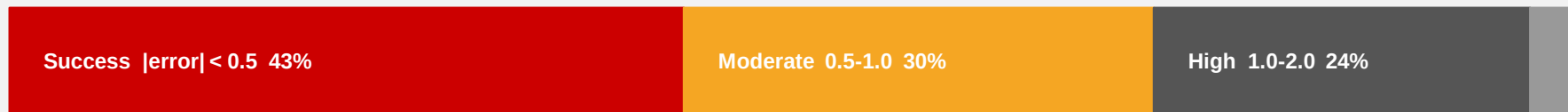
Criterion	Item-Based CF	SVD	Winner
RMSE	0.9720	0.9273	SVD
MAE	0.7527	0.7245	SVD
MAP@10	0.6720	0.6926	SVD
Train Time	~0.1s	~0.4s	SVD
Interpretability	High	Low	Item-CF
Cold Start	Poor	Poor	Tie

SVD wins on all quantitative metrics | Item-CF wins on interpretability and explainability

Performance Metrics Summary

SVD RMSE	SVD MAE	SVD MAP@10	CF RMSE
0.9273	0.7245	0.6926	0.9720
Best prediction accuracy	Avg. 0.72 star error	Ranking quality score	Item-Based CF baseline

Prediction Error Distribution (SVD)



Top-10 SVD Recommendations — Sample User

1	The Shawshank Redemption 1994	★ 4.68	6	Forrest Gump 1994	★ 4.45
2	Schindler's List 1993	★ 4.61	7	LOTR: Return of the King 2003	★ 4.41
3	The Godfather 1972	★ 4.57	8	Goodfellas 1990	★ 4.38
4	The Dark Knight 2008	★ 4.52	9	Fight Club 1999	★ 4.35
5	Pulp Fiction 1994	★ 4.48	10	Silence of the Lambs 1991	★ 4.31

Cold Start: New users receive Bayesian popularity-weighted fallback recommendations (e.g. Toy Story, Finding Nemo)

Key Takeaways

- SVD outperforms Item-CF: RMSE 0.93 vs 0.97
- Item-CF offers interpretable recommendations
- High sparsity favours latent factor models
- Cold-start needs a hybrid fallback strategy
- 43% of SVD predictions within 0.5 stars of truth

Future Improvements

- Neural CF with embedding layers
- Two-tower retrieval + ranking
- Hybrid SVD + content-based
- Temporal decay weighting
- Online A/B testing framework